A Tour of Multiple View Geometry

Multiple View Geometry in Computer Vision, chapter 1

Lesson t3.1 · Hartley, Zisserman

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Why a robot engineer needs this chapter

- A drone, a quadruped and a tracked robot all turn images into geometry
- Which parts of the world can a set of images determine, and which not?
- The answer is a group of transformations, not a vague 'noise'
- A monocular drone loses metric scale; a quadruped turning in place breaks F
- Bundle adjustment and VIO front ends are built on this vocabulary

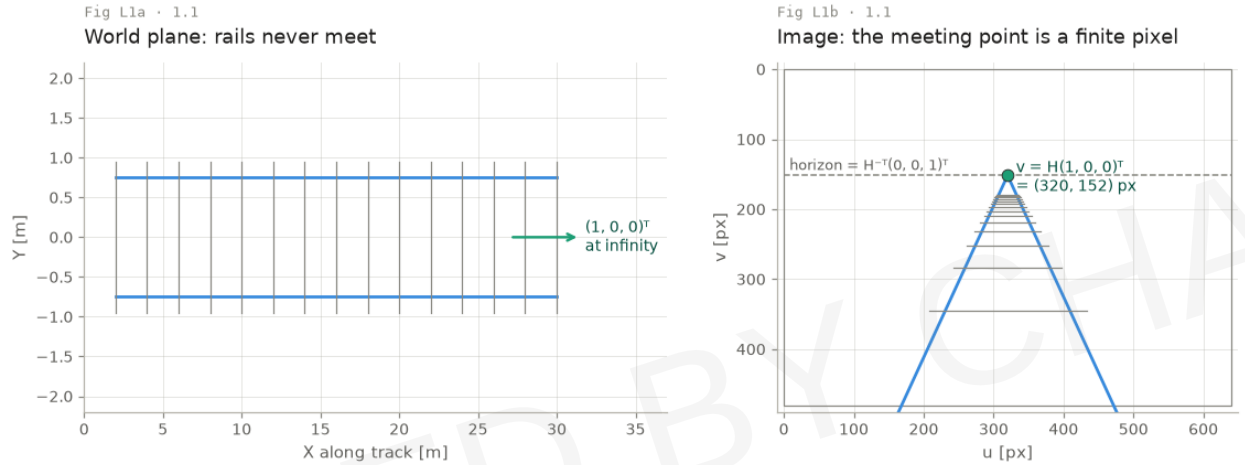
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Roadmap

1. 1.1 Points, infinity, the ladder
2. 1.2 Camera as a 3×4 matrix, calibration as a conic
3. 1.3 Two views: a 4×4 ambiguity
4. 1.7 Climbing back to Euclidean
5. Skims: trifocal, n views, auto-calibration

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1.1 Parallel rails meet in the image



- In the world plane the rails never meet
- In the image they meet at v = (320, 152) px
- v is the image of the direction $(1, 0, 0)^T$

Rails 1.5 m apart, camera 1.6 m up, pitched 10° down

Parallel lines meet at a finite pixel: the image of a point at infinity

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1.1 Homogeneous coordinates and X′ = HX

$$(kx,\ ky,\ k)^\top \sim (x,\ y,\ 1)^\top \mapsto (x,\ y), \quad k \neq 0$$ p. 2

$$\mathbf{X}' = \mathrm{H}_{(n+1)\times(n+1)}\mathbf{X}$$ p. 3

k arbitrary non-zero scale

(x, y, 0)ᵀ point at infinity

H non-singular, up to scale

X, X′ homogeneous (n+1)-vectors

- $(x, y, 0)^\top$ is a point at infinity: a direction
- H non-singular, defined up to scale: 8 DoF for 3×3, 15 for 4×4
- Euclidean and affine maps keep infinity at infinity; a general H does not

A point at infinity is a direction, and H need not keep it there

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Predict: where does a far landmark go?

- A drone sees a landmark at (X, Y, Z, T) with T → 0
- Which parametrization would you put in a filter state?

? **Which landmark representation stays finite and well conditioned as T → 0?**

A (X/T, Y/T, Z/T)

B (X, Y, Z, T)

C Both of them

D Neither of them

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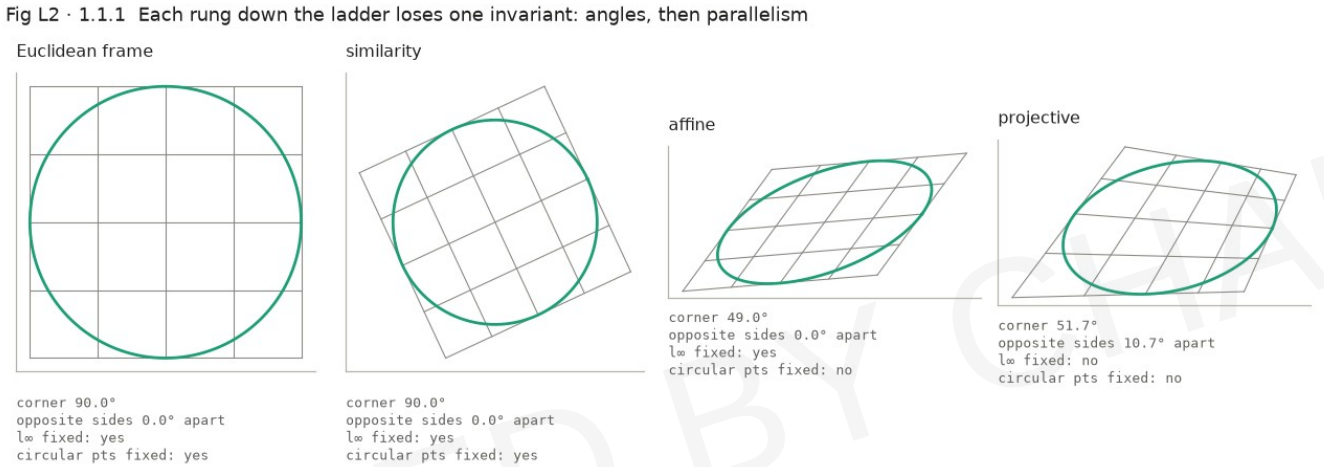
The idea of the ladder

Projective, affine and Euclidean geometry differ in what they keep pinned at infinity

Pin a line at infinity for parallelism; pin the circular points for angles

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1.1.1 The ladder: projective, affine, Euclidean



- Similarity: corner 90.0°
- Affine: opposite sides 0.0° apart, corner 49.0°
- Projective: corner 51.7°, opposite sides 10.7° apart

A square and its inscribed circle through four maps

> **Each rung you drop loses angles first, then parallelism**

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Three geometries, one square

Euclidean / similarity

- Keeps lengths up to scale and angles
- Corner stays 90.0°
- Circular points fixed

Affine

- Keeps parallelism
- Opposite sides 0.0° apart
- Corner becomes 49.0°
- Line at infinity fixed

Projective

- Keeps lines and incidence only
- Opposite sides 10.7° apart
- Corner becomes 51.7°
- Line at infinity moves

Read the numbers left to right: each geometry keeps less than the one before

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1.1.1 Circles and the circular points

$$(x - aw)^2 + (y - bw)^2 = r^2 w^2$$ p. 5

$$x^2 + y^2 = 0, \quad w = 0 \quad \Rightarrow \quad (1, \ \pm i, \ 0)^\top$$ p. 5

(a, b) circle centre

r radius

w homogeneous coordinate

I, J circular points $(1, \pm i, 0)^\top$

- $1 + i^2 = 0$, so I and J lie on every circle, whatever a, b, r
- A conic has 5 DoF; a circle already passes through I and J

Every circle passes through the two circular points I and J

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1.1.1 The absolute conic and perpendicularity

$$\Omega_\infty : \; X^2 + Y^2 + Z^2 = 0, \quad T = 0$$ p. 6

$$\mathbf{d}_1^\top \, \Omega_\infty \, \mathbf{d}_2 = 0$$ p. 6

Ω∞ absolute conic on the plane at infinity

$\mathbf{d}_1, \mathbf{d}_2$ two direction vectors

T fourth homogeneous coordinate

- Every sphere meets the plane at infinity in Ω∞; no real points
- Conjugate directions are perpendicular; in a Euclidean frame Ω∞ = I
- The general angle formula is (3.23), p. 82

Perpendicular directions are conjugate with respect to Ω∞

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Predict: how many points fix a circle? A conic?

- A general conic has 6 coefficients up to scale
- A circle must pass through the circular points I and J

? **How many ordinary points fix a general conic, and how many a circle?**

A 5 and 3

B 6 and 4

C 5 and 2

D 4 and 3

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1.2 The camera centre is what matters

Fig L3a · 1.2, Fig. 1.1c

Same centre: both planes cut the same rays

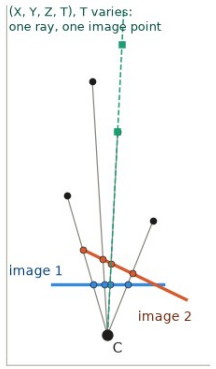
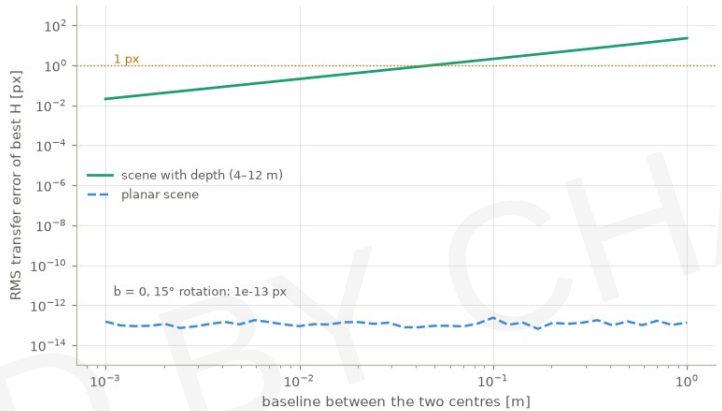


Fig L3b · 1.2, Fig. 1.1c–e

One H fits both views only if C stays put or the scene is flat



- Planar scene: at most 2.4×10^{-13} px at any baseline
- Scene 4–12 m deep: 1 px at 0.048 m, 2.1 px at 0.1 m
- 22.75 px at 1 m

Same-centre views (left); homography fit error vs baseline (right)

A shared centre gives a homography; a moved centre needs a plane

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How far can the centre move before one H fails?

0.048

m of baseline until the fit error reaches 1 px

2.1

px of fit error at 0.1 m

22.75

px of fit error at 1 m

- Scene 4–12 m deep, focal length 500 px
- Planar scene: at most 2.4×10^{-13} px at any baseline

Nearly the same centre is a tight condition once the scene has depth

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1.2 The camera is a 3×4 matrix

$$\mathrm{P} = [\, \mathrm{I}_{3\times 3} \mid \mathbf{0}_3 \,] \qquad \text{p. 7}$$

$$(x,\ y,\ w)^\top = \mathrm{P}_{3\times 4}\,(X,\ Y,\ Z,\ T)^\top, \quad \operatorname{rank}\mathrm{P} = 3 \qquad \text{p. 7}$$

$$H = [\, \mathbf{p}_1\ \mathbf{p}_2\ \mathbf{p}_4 \,] \ \text{ for the plane } Z = 0 \qquad \text{p. 7}$$

P 3×4 camera matrix

(X, Y, Z, T) world point, homogeneous

(x, y, w) image point, homogeneous

H plane homography, 3×3

$\mathbf{p}_1$ … $\mathbf{p}_4$ columns of P

- P = [I | 0] forgets where along the ray a point was
- The centre C satisfies PC = 0; rank 3 makes it unique

Any rank-3 3×4 matrix is a camera; the centre C satisfies PC = 0

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Predict: does a rotation need depth?

- A quadruped turns 15° in place; the scene is 4–12 m deep
- Fit one homography between the two images

? **After a pure rotation, does one homography fit the images of a deep scene?**

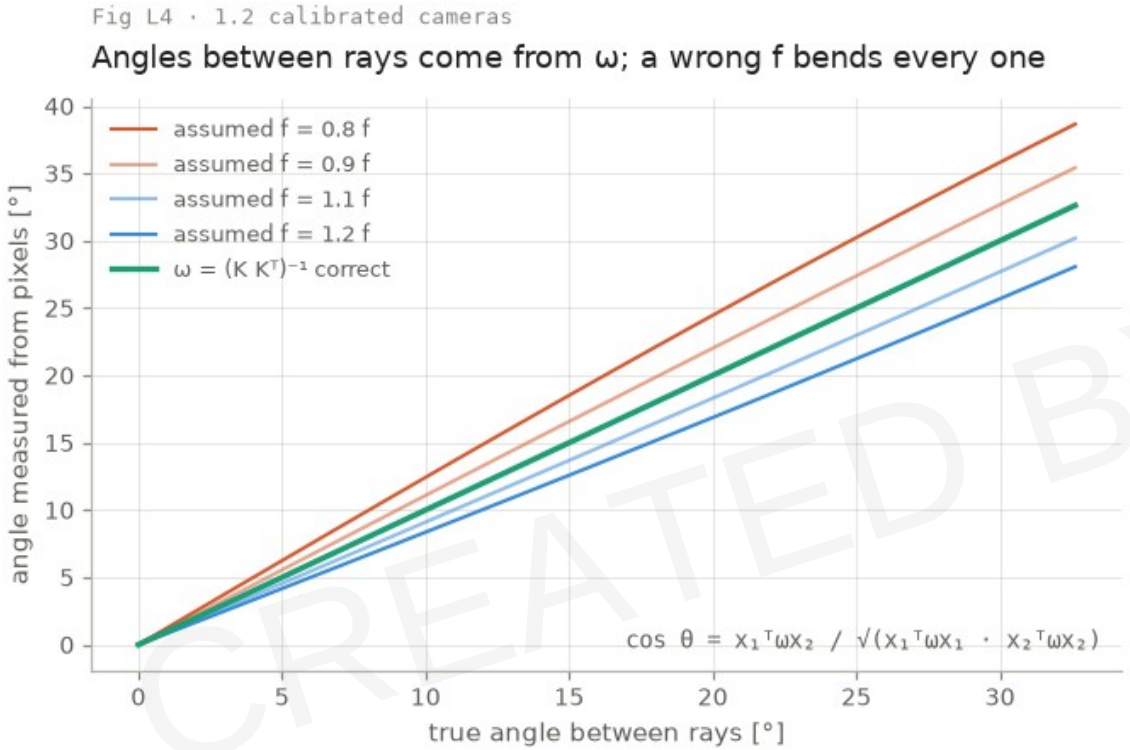
A Yes, the error is near zero

B No, the error grows with depth

C Only if the scene is a plane

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1.2 Calibration as a conic: ray angles



Angle error against pixel position for a wrong focal length

- Correct f gives the true angle: 32.62° at the image edge
- f 20% short reads 6.04° high
- f 20% long reads 4.55° low

A wrong focal length hurts most at the image edge

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1.2 The angle between two rays

$$\cos\theta = \frac{\mathbf{x}_1^\top \omega \mathbf{x}_2}{\sqrt{(\mathbf{x}_1^\top \omega \mathbf{x}_1)(\mathbf{x}_2^\top \omega \mathbf{x}_2)}} \qquad (8.9)$$

$$\omega = (\mathrm{K}\mathrm{K}^\top)^{-1} \qquad \text{Result 8.17}$$

x_1, x_2 homogeneous image points

K calibration matrix, upper triangular

ω image of the absolute conic, 3×3

θ angle between the two rays

- x_1, x_2 homogeneous pixels; ω 3×3 symmetric; K upper triangular
- Example 1.1: vanishing points then right angles give Euclidean (Fig. 1.2)

Calibration is knowing ω; it turns pixels into ray angles

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Predict: size of the error from a wrong f

- Image 640 × 480 px, f = 500 px, true angle at the edge 32.62°
- The calibration file has f set 20% short

? **The focal length is 20% short. How does the edge angle read?**

A High, by about 6°

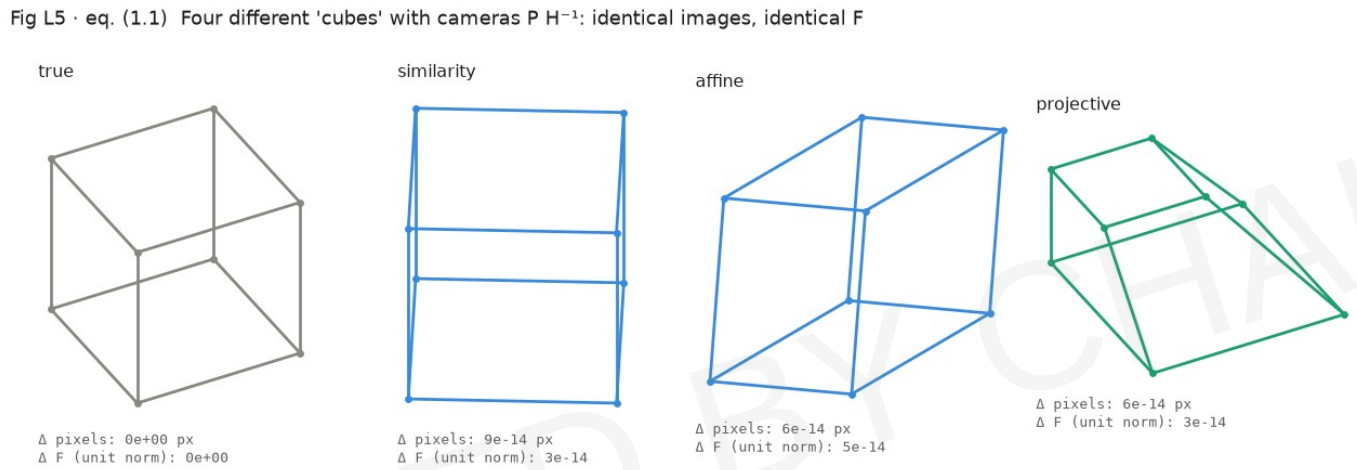
B Low, by about 6°

C No change at the edge

D High, by about 1°

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1.3 Two views: a cube and its look-alikes



- Largest pixel change: 8.5×10^{-14} px
- Largest change in unit-norm F: 4.7×10^{-14}
- The images cannot tell the cubes apart

Cameras $P H^{-1}$ and points $H X$ give the same images

Images alone cannot tell a cube from its projective look-alikes

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1.3 Projective ambiguity and the epipolar constraint

$$\mathrm{P}_j\mathbf{X}_i = (\mathrm{P}_j\mathrm{H}^{-1})(\mathrm{H}\mathbf{X}_i) \qquad (1.1)$$

$$\mathbf{x}_i'^{\top}\,\mathrm{F}\,\mathbf{x}_i = 0, \quad \operatorname{rank}\mathrm{F} = 2 \qquad \text{p. 11}$$

P_j camera j, 3×4

H any non-singular 4×4

X_i scene point i

F fundamental matrix, 3×3, rank 2

x_i, x'_i matched image points

- F has 9 entries: 9 − 1 (scale) − 1 (det F = 0) = 7
- So at least seven matches, outside critical configurations (p. 11)

F has 7 degrees of freedom; the cameras stay free up to H

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1.3 The four-step outline (p. 12)

1 (i) Equations

One linear equation in F per match, solved properly in ch. 11

2 (ii) Solve for F

Ch. 11; outliers are handled in §4.7

3 (iii) Cameras

Cameras P, P′ from F, §9.5, p. 253

4 (iv) Triangulate

A point X_i for each match, ch. 12

The outline is a map of later chapters, not an implementation

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1.4–1.6 Skim: three views, n views, transfer

- Three views: trifocal tensor, 27 entries; 4 equations per triplet
- 7 triplets solve it linearly, 6 non-linearly (p. 13)
- n views: affine factorization by SVD; bundle adjustment needs a start
- Transfer $x'' = P''X$: triangulate from two views, then project
- With a plane or a pure rotation, one homography suffices

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Climbing from projective to metric

1 Projective

Cameras and points known only up to a 4×4 H, eq. (1.1)

2 Affine

Send π_∞ to $(0, 0, 0, 1)^T$ with H_A; parallel edges become parallel

3 Metric

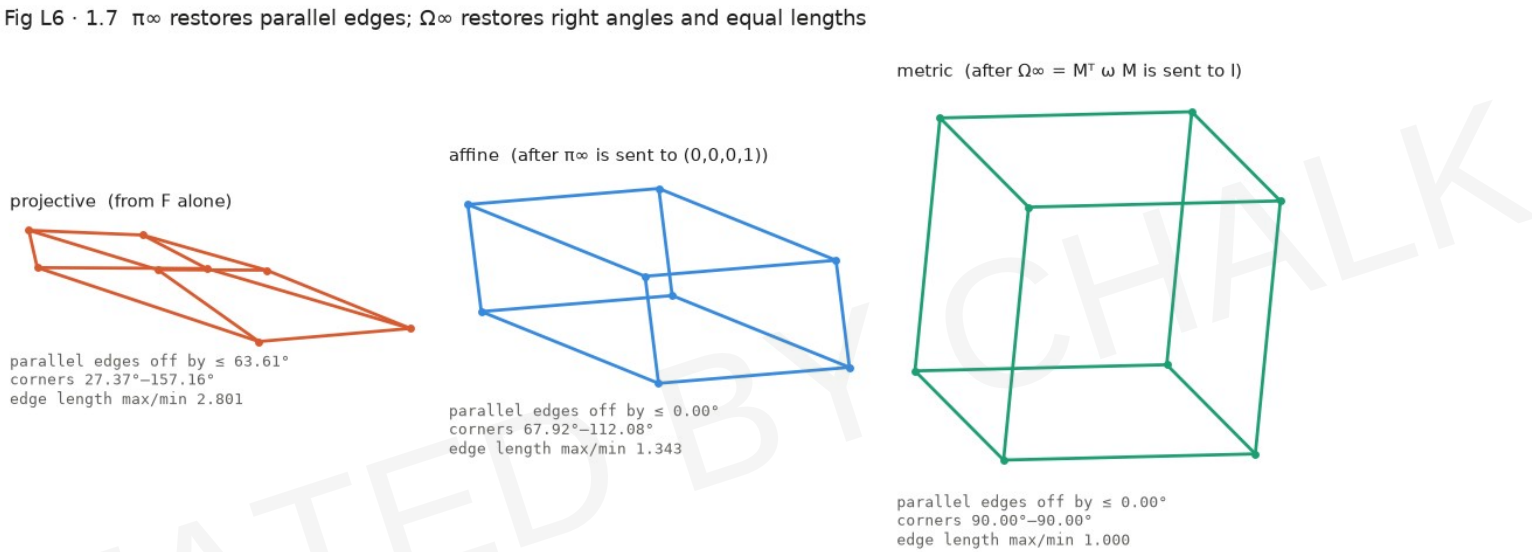
Send Ω_∞ to I with H_M; corners become right angles

- Remaining freedom: a similarity, so no absolute scale

Each step pins one object at infinity and removes one kind of ambiguity

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1.7 Climbing back to Euclidean



- π∞ to $(0, 0, 0, 1)^T$: worst parallel-edge angle 63.61° → 0.00°
- Corners then 67.92°–112.08°; $\Omega\infty = M^T\omega M$ to I: corners 90.00°

A projective reconstruction of a cube upgraded in two steps

Two steps remove the ambiguity: plane at infinity, then absolute conic

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1.7 The two upgrade steps

$$\Omega_\infty = M^{-\top} \omega M = L L^{\top}$$ §10.4

Ω∞ absolute conic

ω image of the absolute conic

M left 3×3 block of a camera

L Cholesky factor

- H_A = [I | 0; π^T] sends π∞ to $(0, 0, 0, 1)^T$ (affine)
- H_M = [L^T 0; 0 1] makes Ω∞ = I (metric); L is the Cholesky factor
- A global similarity remains: scale is not recovered

What is left after calibration is a similarity: shape without scale

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1.8–1.10 Skim: auto-calibration and the rewards

- No calibration knowledge: projective is the limit (p. 17)
- Constant calibration: find π_∞ first (hard), then ω linearly (easy)
- Square pixels: 2n rays; four images fix it up to a finite set
- 3D models: projective, auto-calibration, dense matching along epipolar lines
- Video augmentation needs the Euclidean camera motion

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Traps: which rung did the sensor setup leave open?

Symptom	Cause	Try
F flips between frames	Rotation or planar scene	Fit H and F, pick by score
Parallels converge	π_∞ unknown	Calibrate K; impose parallels
Right angles wrong	Ω_∞ (ω, K) wrong	Recalibrate f, aspect, skew
Bearings off at the edge	Wrong f in ω	Recalibrate
Map size drifts	Scale unobservable	IMU, stereo, scale prior
BA diverges	Bad start or free gauge	Initialize; anchor pose and scale
Panorama ghosting	Centre moved	Rotate about the centre
Bird's-eye warps	Stale horizon	Update H from attitude

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Summary: claims to defend

- Points at infinity are ordinary points once coordinates are homogeneous
- Projective, affine and Euclidean differ by what they pin at infinity
- A camera is a rank-3 3×4 matrix; the centre decides when a homography fits
- Uncalibrated matches fix a scene only up to a 4×4 H, eq. (1.1)
- Calibration is knowing ω, equivalently Ω_∞, and removes all but a similarity

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What comes next

- Ch. 2–3: lines, conics, planes and Ω_∞ become definitions; angle formula (3.23)
- Ch. 6–8: cameras as $K[R \mid t]$, and $\omega = (KK^T)^{-1}$
- Ch. 9–12: two-view geometry, F, cameras from F, triangulation
- Ch. 18–19: n-view methods and auto-calibration
- Robot side: graph SLAM, FAST-LIO2, and monocular VIO scale

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